

— 2026 —

强化学习暑期学校

2026 SUMMER SCHOOL ON REINFORCEMENT LEARNING

📅 2026.07 | 📍 中国·上海

Course Project Plan



At a Glance

A three-week, team-based mini research project. Core RL is the default path; embodied and LLM/agentic streams are capped exceptions.

~94

Students

~40

Teams

1–3

Per team

3 wks

Jul 6 – 24

Jul 23

Poster session

Each team delivers

- Half-page proposal
- Baseline / minimal reproducible result
- One or two extensions, ablations, or a conceptual analysis
- Mini-paper / report (4–6 pages)
- Poster + 3–5 minute explanation
- Reproducibility package (code, command)



Objectives & Core Principle

1

Internalize core RL

Understand core RL ideas through concrete, controlled investigation.

2

Connect to research

Let motivated students link those ideas to current directions — within clear three-week bounds.

CORE PRINCIPLE

The poster and the project must state what the reinforcement learning question is, not only that a model scored higher.

Judged on question clarity, connection to core RL, reproducibility, and honest interpretation — not engineering scale or benchmark scores.



Three Project Streams

 Core RL

> 25 teams

Default, recommended, and clear majority. Every team completes a clean, reproducible mini study.

Compute Local / CPU first / Shared 4090

 Embodied / Sim RL

< 10 teams

Advanced exception for prepared teams; strict scope control and prior approval.

Compute Shared 4090, pre-approved

 LLM / Agentic RL

<10 teams

Exploratory exception only: RLHF, preference learning, reward modeling, LLM agents.

Compute H100 / H200, small-scale



Core RL Topics: Alberta Plan

Close to the course and to Rich's directions over the past decade.

- **Streaming/Online/Continual learning**
single-stream updates without a replay buffer
- **Average reward vs discounting**
reward centering / reward-shift robustness
- **Reward hypothesis**
reward is enough or not? How and where to get reward?
- **Step-size adaptation**
IDBD / TIDBD on non-stationary tasks
- **Options & subtasks**
eigen- vs bottleneck vs reward-respecting (Four Rooms)
- **Agent state and representation**
construction from a stream under partial observability
Tile coding? Or other way to represent observation state
- **Planning**
Dyna, prioritized sweeping, expected vs sample models
- **n-step & eligibility traces**
sensitivity in n and λ ; representation choices
- **GVPs & predictive knowledge**
question functions and “nexting” predictions
- **Off-policy stability**
Baird's counterexample; TD vs gradient- and emphatic-TD

The Alberta Plan for AI Research: <https://arxiv.org/pdf/2208.11173>



Recommended Starter Environments

■ Tabular & chains

random walk, gridworld, windy gridworld, cliff walking, Four Rooms

■ Classic control

Acrobot, CartPole, Mountain Car, Pendulum

■ Off-policy / function approx.

Baird's counterexample, tile-coding prediction/control, Mountain Car

■ GVF / predictive

small Markov chains, gridworld, or synthetic sensor streams

■ Continuing / average-reward

two-loop or cycle MDP, continuing gridworld, access-control queuing

■ Non-stationary / continual

drifting or permuted targets, non-stationary bandit or gridworld



Three-Week Timeline

When	Stage	Student deliverable
Jul 6	Start & team formation	Team formed; core RL direction selected
Jul 7–9	Frame the question	Half-page proposal. Question, setting, and expected behavior
Jul 10–15	Build & observe	Minimal study and initial observations
Jul 16	Project checkpoint	Brief update; continue, simplify, or refocus
Jul 20–21	Investigate & explain	Focused investigation and interpretation
Jul 22	Poster draft	Poster draft and main takeaway
Jul 23	Poster session	Poster, brief explanation, and project materials

Core RL is the default. Keep the project clear enough to understand, run, and explain within three weeks.



Deliverables & Checkpoints

Proposal

by Jul 9

- Question & hypothesis
- Environment & baseline
- Metric & minimal result
- Compute need & fallback

Final package

Jul 23–24

- Mini-paper (4–6 pp) + poster
- Code + exact run command
- Seed(s) + commit id + result path
- 1-page reproduction note + reflection

No baseline by Jul 16 → simplify the topic, change environment, or convert to a conceptual analysis with a small reproducible example.



Compute Support Plan

An upper-bound plan, not an entitlement. Core RL runs local / CPU; shared GPU is reserved for approved simulation and LLM work, final reproductions, and failures.

RTX 4090

Up to 1000 GPU hour per approved core-RL or embodied team

H100 / H200

LLM stream (< 10 teams): up to 1000 H200 hour each

Governance

No large-model training from scratch

No blind grid search



TA Support & Contacts

Project lead 1

Ge Xinmu @葛馨木

Core RL: function approximation, off-policy, deadly triad, n-step, traces, GVFs, options

Project lead 2

Wang Zeyuan @王泽源

Compute platform; embodied sim RL; Gymnasium / MuJoCo / Isaac; environment smoke tests

Project lead 3

Bai Bizhe @白碧哲

LLM / agentic RL; LoRA; preference optimization; H100/H200 coordination; reproducibility

Poster lead

Bai Bizhe @白碧哲

Poster session plan, schedule, sectioning, and feedback process

Archive owner

Xu Guangjun @徐光君

Poster / code / result collection; excellent-project candidates; archive directory

Use role-based routing first. Freeze the owner list before the first project clinic; leads jointly maintain the master project table.

More Supports are also available:

Jun Luo @罗军 from Openmind

Ying Wen @温颖 from SII/SAI@SJTU



Poster Session · July 23

A focused poster walk-around with brief wrap-up.

14:00–14:10

Opening & section explanation

14:10–15:10

Poster walk-around, round 1 — students explain; Rich, teachers, TAs, Openmind, visitors circulate

15:10–15:45

Rich / Openmind / faculty focused walk-around

15:45–16:00

Wrap-up & collection

Every poster & mini-paper opens with its RL question. Include: why it matters · environment · method & baseline · result or failure · interpretation · next step · reproducibility.